4 – Execute the Burgwald Data Pipeline

Clone · Configure · Run · Verify

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Goals

- Clone the course repository and open the RStudio project.
- Run all core pipeline scripts **unchanged** on your own machine.
- Verify that raw data, processed data, and predictor stacks are created.
- Document any problems that prevent full reproducibility.

Written output...

Please export all written answers as a PDF and save it in the project directory `docs/` using the filename convention `<Name1_NameN>_ws-04.pdf`. This helps you maintain a clean and consistent documentation of your work.
You are welcome to use ChatGPT for wording support, but the important part is that you genuinely understand the tasks and the underlying concepts — the AI is only meant to reduce typing effort, not replace your comprehension.

! How to use ChatGPT for this assignment

You may use ChatGPT **only** to refine the wording of your own ideas.
Please follow this workflow:

1. **Write your own bullet points first** (even rough or incomplete).
2. Paste your bullets into ChatGPT and ask for **clarification, shortening, or better phrasing**.
3. ChatGPT will only help with **language and structure**, not with content.
4. If you are unsure about something, write: *"I'm not sure about X"* — ChatGPT may then give **hints or guiding questions**, but **no solutions**.
5. Answers must remain **ultra-short and clear** (1–3 bullets or 1–2 sentences).
6. Do **not** ask ChatGPT for full answers; the goal is that **you understand the task**, not that AI solves it.

This ensures that the work remains genuinely yours while reducing unnecessary typing.

Repository

Clone this repository locally and open the `.Rproj`:

https://github.com/gisma/LV-19-d19-006-25-R.git

Task 1 – Run the setup

Script: `01_setup-burgwald.R`

- Open the script.
- Run it from top to bottom **without any modifications**.

- After successful execution, the following must exist:

```
data/raw/AOI_Burgwald/
data/processed/aoi_burgwald.gpkg
src/01-fun-data-retrieval.R  (loaded without error)
```

Minimal check (optional in console):

```
fs::dir_ls("data", recurse = TRUE)
```

Task 2 – Run generic data retrieval

Script: `01-data-retrieval.R`

- Run the script and change the time-peroid to something like 1 month.
- Afterwards, the following data must physically exist (content not relevant, existence is):
  - DEM (GeoTIFF)
  - OSM-by-key files
  - CLC raster / GeoPackage
  - DWD station data in `data/raw/dwd-stations/`
  - processed DWD files in `data/processed/dwd/`

Suggested quick check:

```
fs::dir_ls("data", recurse = TRUE)
```

Task 3 – Test external API (SAGA)

Script: `01-04-1-SAGA.R`

- Adjust the SAGA binary path if needed (only this, nothing else).
- Run the script.
- At least **one new terrain raster** (e.g. slope, aspect) has to be created.
- Note the file name and location.

Task 4 – Sentinel-2 via CDSE

Script: `01-2-CDSE-sentinel-data-retrieval.R`

- Run the script unchanged.
- Expected result (folder may vary slightly, adapt to repo):

```
data/raw/s2/
      <Sentinel-2 scenes or COG references>
```

- Count/list how many items were downloaded or referenced.

Task 5 – Sentinel-2 via gdalcubes

Script: `01-3-gdalcubes-sentinel-data-retrieval.R`

- Run the script unchanged.
- Expected output:

```
data/predictor/
       <NetCDF file>   # e.g. s2_predictor_stack_summer.nc
```

- Confirm that the NetCDF file exists and can be opened, e.g.:

```
stars::read_stars("data/predictor/<your-file>.nc")
```

Task 6 – Pipeline check

Run this final check:

```
list.files("data/raw", recurse = TRUE)
list.files("data/processed", recurse = TRUE)
list.files("data/predictor", recurse = TRUE)
```

Verify that all three stages exist on your system:

- RAW DATA
- PROCESSED DATA
- PREDICTOR STACK(S)

If any stage is missing, re-run the corresponding script or document the error.

Task 8 – Short reproducibility log

Write **one short paragraph** (max. 5–6 sentences) for your own log:

- Did the full pipeline run on your machine?
- Which scripts worked without issues?
- Where did you have to change paths or credentials (e.g. SAGA, CDSE)?
- Which errors (if any) stopped the pipeline?

You will use this later to compare reproducibility between systems.

Expected final structure

The expected directory tree at the end of this worksheet should be more or less like this:

```
data/
  processed
      dwd-stations
      osm_by_key
      radolan-rw
      relief_1m
  raw
       AOI_Burgwald
          clc
          dem
          osm_by_key
       clc5_2018_copernicus
          Results
       dgm1_burgwald
       dgm1_coelbe
       dgm1_gemuenden
       dgm1_haina
       dgm1_lahntal
       dgm1_muenchhausen
       dgm1_rauschenberg
       dgm1_rosenthal
       dgm1_wetter
       dgm1_wohra
       dwd-stations
          unzipped_10min_precip
          unzipped_5min_precip
          unzipped_hourly_
              precipitation
              wind
          unzipped_hourly_precipitation
          unzipped_hourly_wind
       radolan-rw
       u2018_clc2018_v2020_20u1_raster100m
           DATA
               French_DOMs
           Documents
           French_DOMs
               Legend
           Legend
           Metadata  predictor/
  predictos
 src/
```

```
outputs/
     figures/
 docs
 metadata
 renv
```

Take-home summary

This pipeline forms the foundation for all later modelling tasks in the course.By running it successfully, you ensure that:

- all Burgwald base layers (DEM, OSM, CLC, DWD) are available,
- all Sentinel-2 sources (CDSE + gdalcubes) are standardised and reproducible,
- the project contains a clean, FAIR-aligned data structure, and
- downstream scripts (classification, predictors, modelling) can run without modification.

In short: if this pipeline runs, every later analysis in the course becomes plug-and-play.